AI Candidate Ranking System

INDIA RUNS — Redrob AI × Hack2skill Hackathon

An AI-powered system that ranks candidates the way a great recruiter would —

by understanding who genuinely fits the role, not by matching keywords.

Track 1: AI-Powered Candidate Ranking

The Problem

Recruiters review hundreds of profiles and still miss the right person.

Not because talent isn't there — but because keyword filters can't see what matters.

- Keyword matching misses context and intent
- Behavioral signals (urgency, activity) are ignored
- No explainable scoring — recruiter can't trust black-box ranking
- Indian hiring platforms need India-specific understanding

Our Solution

Capability	Approach
Understands job context	Semantic embeddings (sentence-transformers)
Goes beyond keywords	Synonym-aware skill + sector matching
Full candidate picture	8 scoring dimensions
Explainable scores	Transparent breakdown per candidate
India-first	Built for Indian education, languages, sectors
Flexible	Works with any candidate data source

System Architecture

Job Description → **JD Parser** → Extract skills, sector, location, experience

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Embedding Engine (sentence-transformers all-MiniLM-L6-v2)
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Similarity Computation (cosine similarity — 384d embeddings)
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Hybrid Scorer (8 weighted dimensions)
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Ranker → **Ranked Candidate Shortlist**

Scoring Dimensions (Weighted Hybrid)

Dimension	Weight	What It Measures
Semantic Similarity	30%	Embedding cosine similarity — understands context
Skills Match	20%	Direct + synonym-aware skill matching
Experience Match	15%	Years vs requirements; handles fresher/senior
Qualification	10%	Education level alignment
English Proficiency	5%	No → Basic → Good → Fluent
Behavioral Signals	10%	Urgency, resume, verification, activity
Location Match	5%	City/location alignment
Sector Match	5%	Industry sector relevance

Hybrid Score = 0.30×Semantic + 0.20×Skills + 0.15×Experience + 0.10×Qualification + 0.05×English + 0.10×Behavioral + 0.05×Location + 0.05×Sector

Semantic Understanding Beyond Keywords

Why embeddings beat keyword matching:

- 'Delivery Executive' matches 'Bike Rider + Packing' semantically — even with zero keyword overlap
- Model: sentence-transformers **all-MiniLM-L6-v2** (384-dim normalized embeddings)
- Cosine similarity in embedding space captures role relevance
- Handles synonyms: 'security guard' ↔ 'watchman', 'sales' ↔ 'business development'

```
# Example: job text → 384-dim vector
job_vec = model.encode("Delivery Executive in Mumbai")
cand_vec = model.encode("Bike Rider, Packing, Mumbai")
similarity = cosine_similarity(job_vec, cand_vec) # High match
```

Datasets

Dataset	Source	Size	Type
WorkIndia	Real candidate profiles	100	Blue/grey collar — skills, experience, behavioral signals
Role Radar Profiles	438 synthetic + 202 real	640	Professional profiles with roles, skills, domains
Scraped Jobs	Real Indian job postings	2,500	Full JD with title, description, location, seniority
Gold Labels	Expert-annotated pairs	108	Ground truth for evaluation

WorkIndia candidates include: skills, experience, qualification, sectors, languages, English level, location, urgency flag, resume flag, mobile verification, last seen timestamp, hot lead status.

Sample Ranking Output

Job: Delivery Executive — Mumbai

Rank	Name	Score
#1	Sumit Ghorpade	0.6940
#2	Jeroy Rodrigues	0.6331
#3	Abhishek Suresh Gotad	0.6322
#4	Azman Khan	0.6202
#5	Aalok Jitendra Singh	0.6172

Each candidate includes an **explainable score breakdown**:


```
"score_breakdown": {
"semantic_similarity": 0.6466,
"skills_match": 1.0,
"experience_match": 0.8,
"qualification_match": 0.2,
"english_proficiency": 0.6,
"behavioral_signals": 0.3,
"location_match": 1.0,
"sector_match": 1.0
}
```

Why This Approach Works

• **Beyond keywords** — Semantic embeddings understand role context, not just words

• **Holistic picture** — 8 scoring dimensions cover skills, experience, education, English, behavior, location, sector

• **Explainable** — Recruiters see *why* each candidate ranks where they do, building trust

• **India-first** — Built for Indian platforms (WorkIndia), education levels (10th Pass, 12th Pass), languages (Hindi, Marathi, etc.), and job sectors (delivery, security, sales)

• **Hybrid strength** — Neural embeddings capture nuance; rule-based matching provides reliability

• **Flexible architecture** — Works with any data source; pluggable scoring dimensions

Technology Stack

Component	Technology
Embeddings	sentence-transformers (all-MiniLM-L6-v2)
Similarity	Cosine similarity (numpy)
Scoring Engine	Custom hybrid (Python)
Data Processing	pandas, json
CLI Interface	Python argparse
PDF Generation	reportlab

Why all-MiniLM-L6-v2?

- 384-dim embeddings — good balance of speed and accuracy
- 6-layer transformer — fast inference on CPU
- Multilingual support — handles Indian English and Hinglish
- Lightweight (80MB) — runs on any machine

How to Run

```
# Install dependencies
pip install -r requirements.txt

# Rank WorkIndia candidates for default JD
python src/main.py --source workindia --top-n 10

# Custom job description
python src/main.py --source workindia \
--jd data/job_description_sales.txt --top-n 10

# Use Role Radar dataset
python src/main.py --source jd_dataset \
--jd-index 0 --top-n 10
```

Output: **output/ranked_candidates.json** — ranked list with scores

Future Enhancements

• **LLM-based ranking** — Claude/GPT for deep reasoning on nuanced fit
• **Vector database** — FAISS/Chroma for large-scale retrieval (10K+ profiles)
• **Resume parsing** — PDF/DOCX extraction for direct resume upload
• **Bias detection** — Fairness metrics across age, gender, location demographics
• **Recruiter dashboard** — Streamlit/Gradio UI for interactive review
• **REST API** — FastAPI endpoint for ATS integration
• **Feedback loop** — Recruiter feedback tunes weights over time

Thank You

Built for INDIA RUNS — Redrob AI × Hack2skill

GitHub: github.com/yourusername/candidate-rank-ai

Clean • Modular • Explainable • India-first